ROBERT KOCH INSTITUT

Coronavirus Disease 2019 (COVID-19)

Daily Situation Report by the Robert Koch Institute

20/05/2022- CURRENT STATUS FOR GERMANY

Confirmed cases		7-day incidence (7-di)		DIVI-Intensive care registry as of 19/05/2022 12:15 AM	Vaccination monitoring as of 20/05/2022
Total[1]	active cases[2]	Total population	No. of districts with 7-di > 50/100,000 pop.	Change to previous day for no. of cases currently in ICU	No. of vaccinations reported in last 24h
+48,910 (25,998,085)	-52,100 [ca. 1,333,600]	361.8 cases/100,000 pop.	±0 [411/411]	-32 [873]	1st vaccination: +1,508 2nd vaccination: +3,458 Booster vaccination: +13,560
Hospitalised[1]	Recovered[3]	Hospitalised (all age groups)	No. of districts with 7-di > 500/100,000 pop.	% of COVID-19 occupancy in total number of operable beds[4]	Total no. of people fully vaccinated against COVID-19[5, 6, 7]
+885 (549,836)	+100,800 (ca. 24,526,300)	3.12 cases/100,000 pop.	-21 [59/411]	4.0 %	N1: 64,529,215 N2: 63,074,331 N3: 49,551,387
Deaths[1]		Hospitalised aged 60 years +	No. of districts with 7-di > 1000/100,000 pop	No. of new cases in ICU compared to the previous day	Share of population fully vaccinated against COVID-19
+151 (138,204)		6.94 cases/100,000 pop.	-4 [0/411]	+103	N1: 77.6 % [5] N2: 75.9 % [6] N3: 59.6 % [7]

Numbers in () brackets show cumulative values, numbers in [] brackets show current values. Footnotes can be found in the Annex.

COVID-19 cases are notified to the local public health authorities in the respective districts, in accordance with the German Protection against Infection Act (IfSG). The data are further transmitted through the respective federal state health authority to the Robert Koch Institute (RKI). This situation report presents the uniformly recorded nationwide data on laboratory-confirmed COVID-19 cases transmitted to RKI.

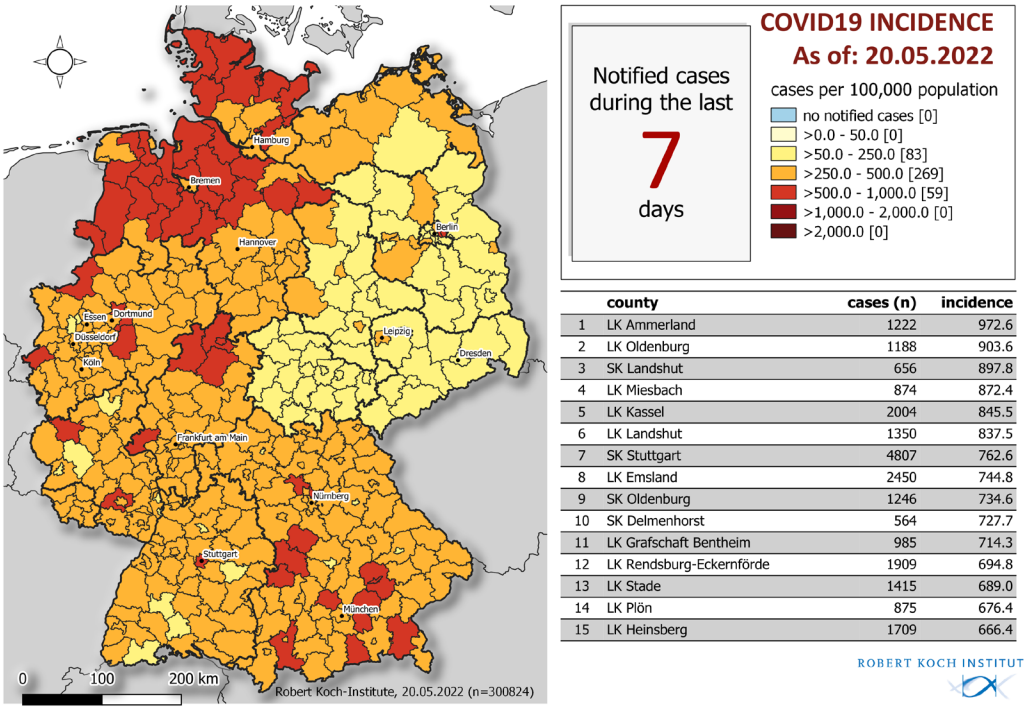
– Changes since the last report are marked blue in the text –

Summary (as of 20/05/2022, 10:00 AM)

- Yesterday, 48,910 new laboratory-confirmed COVID-19 cases as well as 151 new deaths associated with COVID-19 were transmitted to the RKI in Germany. The national 7-day incidence is 361.8 cases per 100,000 population. The 7-day incidence in federal states lies between 545.2 cases per 100,000 population in Schleswig-Holstein and 137.3 per 100,000 population in Thuringia.
- Overall, +885 new hospitalisations with COVID-19 were reported, the 7-day incidence of hospitalised cases is 3.12 per 100,000 population.
- On 19/05/2022 (12:15 AM), 873 COVID-19 patients were in intensive care units (ICU), -32 cases compared to the day before. The COVID-19 adult occupancy as a percentage of all operational adult intensive care beds is 4.0 %.[4]
- Since 26/12/2020, 179,669,929 vaccine doses have been administered in Germany. Overall, 77.6 % of the population in Germany have been vaccinated at least once.[5] 75.9 % have received a complete course of vaccination against COVID-19. [6] 59.6 % have received a booster vaccination.[7]

Epidemiological Situation in Germany (as of 20/05/2022, 0:00 AM)

Since January 2020, a total of 25,998,085 (+48,910) laboratory-confirmed cases of COVID-19 have been reported to and validated by the RKI (Table 1). The geographical distribution of cases of the last 7 days is shown in Figure 1. Please see the COVID-19 dashboard (https://corona.rki.de/) for information on the number of COVID-19 cases by county (local health authority).



	county	cases (n)	incidence
1	LK Ammerland	1222	972.6
2	LK Oldenburg	1188	903.6
3	SK Landshut	656	897.8
4	LK Miesbach	874	872.4
5	LK Kassel	2004	845.5
6	LK Landshut	1350	837.5
7	SK Stuttgart	4807	762.6
8	LK Emsland	2450	744.8
9	SK Oldenburg	1246	734.6
10	SK Delmenhorst	564	727.7
11	LK Grafschaft Bentheim	985	714.3
12	LK Rendsburg-Eckernförde	1909	694.8
13	LK Stade	1415	689.0
14	LK Plön	875	676.4
15	LK Heinsberg	1709	666.4

Figure 1: Number and cumulative incidence (per 100,000 population) of reported COVID-19 cases in Germany by county and federal state (n=300,824, 20/05/2022, 12:00 AM). Cases are usually reported according to the district from which they were transmitted. This usually corresponds to the place of residence. Place of residence and probable place of infection do not have to coincide.

Figure 2 shows the course of the COVID-19 cases per 100,000 population transmitted to the RKI on the last 7 days in each of the federal states and in all of Germany. The values for the 7-day incidence in the federal states range from 545.2 per 100,000 population in Schleswig-Holstein to 137.3 per 100,000 population in Thuringia.

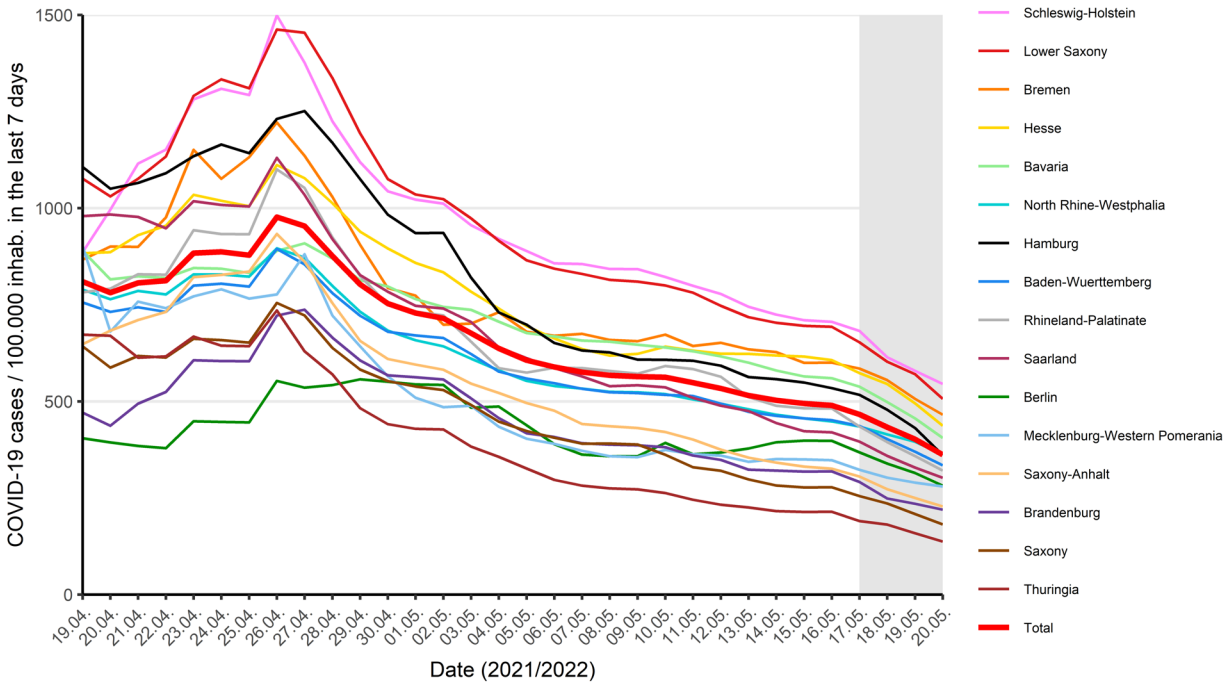


Figure 2: COVID-19 cases/100,000 inhabitants during 7 days in Germany by federal state and reporting date in the health offices (20/05/2022, 0:00 AM). The grey area delineates a range of dates with yet incomplete data, where changes in incidence are likely to occur.

Table 1: Number and cumulative incidence (per 100,000 population) of laboratory-confirmed COVID-19 cases, hospitalizations and deaths for each federal state electronically reported to RKI, Germany (20/05/2022, 12:00 AM). The number of new cases includes cases newly notified to the local public health authority, but also on prior days.

	Cumulative cases			Last 7 days			Cumulative deaths	
Federal State	Number of cases	Number of new cases	Cases/ 100,000 pop.	Cases	Cases/ 100,000 pop.	Hospitalisations/ 100,000 pop.	Number of deaths	Deaths/ 100,000 pop.
Baden-Wuerttemberg	3,634,699	5,252	32,736	37,158	334.7	2.52	16,023	144
Bavaria	4,845,441	7,554	36,875	53,275	405.4	3.92	23,943	182
Berlin	1,037,139	1,637	28,306	10,334	282.0	1.83	4,574	125
Brandenburg	785,100	895	31,018	5,567	219.9	2.17	5,643	223
Bremen	197,528	361	29,043	3,165	465.4	1.47	772	114
Hamburg	574,420	1,389	31,008	6,686	360.9	1.24	2,637	142
Hesse	1,829,345	3,384	29,069	27,491	436.8	3.70	10,067	160
Mecklenburg-Western Pomerania	482,814	646	29,974	4,506	279.7	6.64	2,214	137
Lower Saxony	2,322,752	6,794	29,022	40,556	506.7	2.71	9,206	115
North Rhine-Westphalia	5,202,683	12,408	29,024	64,839	361.7	3.40	25,244	141
Rhineland-Palatinate	1,135,573	2,685	27,708	13,155	321.0	3.56	5,634	137
Saarland	312,447	431	31,753	2,974	302.2	2.64	1,671	170
Saxony	1,500,881	1,155	36,995	7,365	181.5	1.85	15,480	382
Saxony-Anhalt	712,623	916	32,679	4,970	227.9	3.26	5,331	244
Schleswig-Holstein	718,266	2,875	24,675	15,871	545.2	2.99	2,513	86
Thuringia	706,374	528	33,316	2,912	137.3	3.58	7,252	342
Total	**25,998,085**	**48,910**	**31,265**	**300,824**	**361.8**	**3.12**	**138,204**	**166**

Quality checks and data cleaning by the local health departments and federal state health authorities can lead to corrections to cases previously transmitted (e. g. detection of duplicate reports). This can occasionally lead to negative values for the number of new cases.

The precision of the daily reporting of new infections is limited during the weekend and early in the week, because of a reduced level of testing, as well as of reporting and transmission of cases to the RKI (non-mandatory on weekends). Because there are no immediate consequences on the state and federal level, fewer health departments are transmitting data on weekends. Daily fluctuations in case numbers, especially on and just after weekends, should be interpreted with care. In terms of trends, the data are more reliable when comparing week to week. The RKI issues a more detailled weekly report each Thursday [in German].

Estimation of the reproduction number (R), taking into account the reporting delay (Nowcasting)

Figure 3 shows the course of estimated 7-day R-value.

7-day R-value
0.86
(95%-prediction interval: 0.81 – 0.92)

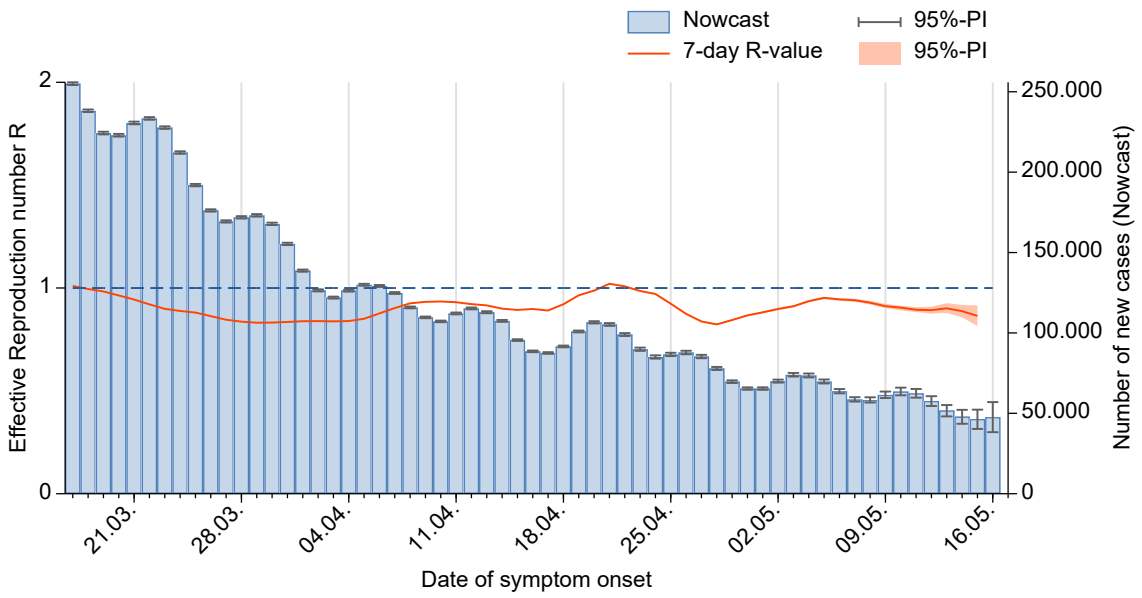


Figure 3: Estimated 7-day R-value (in orange) over the last 60 days, against the background of estimated number of COVID-19 cases according to illness onset (as of 20/05/2022, 12 AM, taking into account cases up to 16/05/2022).

Sample calculations as well as an excel sheet presenting the daily updated R-value can be retrieved under www.rki.de/covid-19-nowcasting. A detailed description of the methodology is available at https://www.rki.de/DE/Content/Infekt/EpidBull/Archiv/2020/17/Art_02.html (Epid. Bull. 17 | 2020 from 23/04/2020).

Annex:

Notes on data collection and evaluation

The data presented in this situation report represent a temporal snapshot. Information on individual cases can be obtained and added by the health authorities in the course of the disease. It is not possible to obtain complete data for all variables.

If necessary, the local public health authorities collect additional information, evaluate reported cases and initiate the necessary infection control measures. In accordance with the Infection Protection Act, the data are transmitted electronically by the local public health authorities to the federal state health authority and from there to the RKI on the next working day at the latest. The data is updated at the RKI daily at 0:00 AM.

Data entry and data transmission can lead to a time lag from the time the case is reported to the local public health authorities until publication by the RKI, so case numbers may deviate from those from other sources.

For the calculation of the incidences, the data of the population statistics of the Federal Statistical Office from 31.12.2020 are used. The calculation of the 7-day incidence is based on the reporting date, i.e. the date on which the local public health authorities became aware of the case and recorded it electronically. For the 7-day incidence, the cases reported on the last 7 days are counted.

On the other hand, the number of cases since the previous day, as shown in the situation report and dashboard, is tied to the date when the case is first published in the RKI's reporting. Thus, due to transmission delays, it may occur that cases with a reporting date more than 7 days ago are still included in the situation report. At the same time, the number of cases since the previous day also may contain cases that were subsequently deleted in data quality checks. Thus, the 7-day incidence cannot be readily calculated from these single-day incidences.

Notes

1 The number of cases since the previous day refers to the date of receipt at the RKI; due to the delay in transmission, cases from previous days may be included.
2 The number of active cases results from the number of transmitted cases minus the deaths and the estimated number of recovered cases.
3 The algorithm for estimating the number of people who have recovered assumes an average duration of illness from onset or hospitalization date on. The true length of symptomatic illness and/or sequelae are usually not available in the reporting system.
4 Proportion of COVID-19 occupancy by adults out of all available adult intensive care beds.
5 The total number of persons vaccinated at least once results from the number of reports coded as first vaccination in the transmitted vaccination data.
6 According to specifications for vaccination data transmission, every second vaccination or first vaccination after recovery is to be transmitted with the code as a complete vaccination. A vaccination with the Janssen vaccine is coded and transmitted as a first vaccination, but is also counted as completed vaccination series.
7 The total number of persons with booster vaccination results from the number of reports coded as booster vaccination in the transmitted vaccination data.